

humanity of the third millenium under assumption of exponential growth

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Abstract

An exponential curve had been matched to the relevant data between the years 1900 and 2021/2023 in these areas: total population, life expectancy, GDP (global total & per capita), energy consumption (global total & per capita).

Results are presented as plots showing the curve and the real data together, as well as plots & tables showing resulting values for the future between years 2000 and 3000.

This is in no way an argument for or against the idea that humanity in the real world will follow a general trend of exponential growth into the future, but rather merely a computation of the values that would result from applying that assumption for use in worldbuilding.

DISCLAIMER AND WARNING

This is not a scientific paper or engineering advice. This document is only intended as creative world-building and has not been peer-reviewed or approved for any practical application. Use of these ideas in real-world designs or other serious application is entirely at the reader's discretion and risk.

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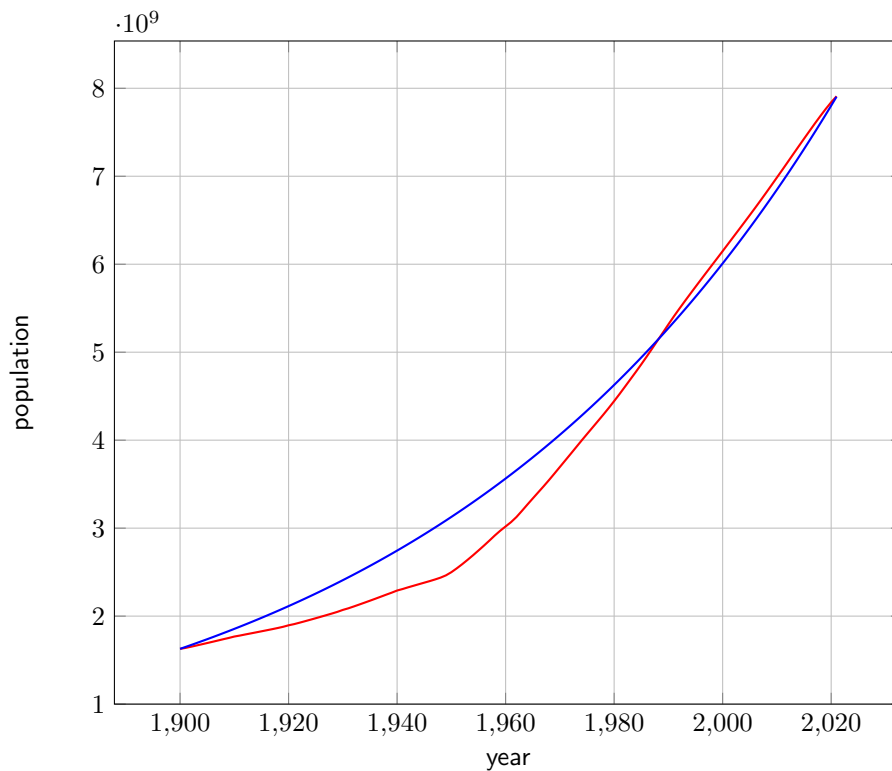
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1 exponential human population growth

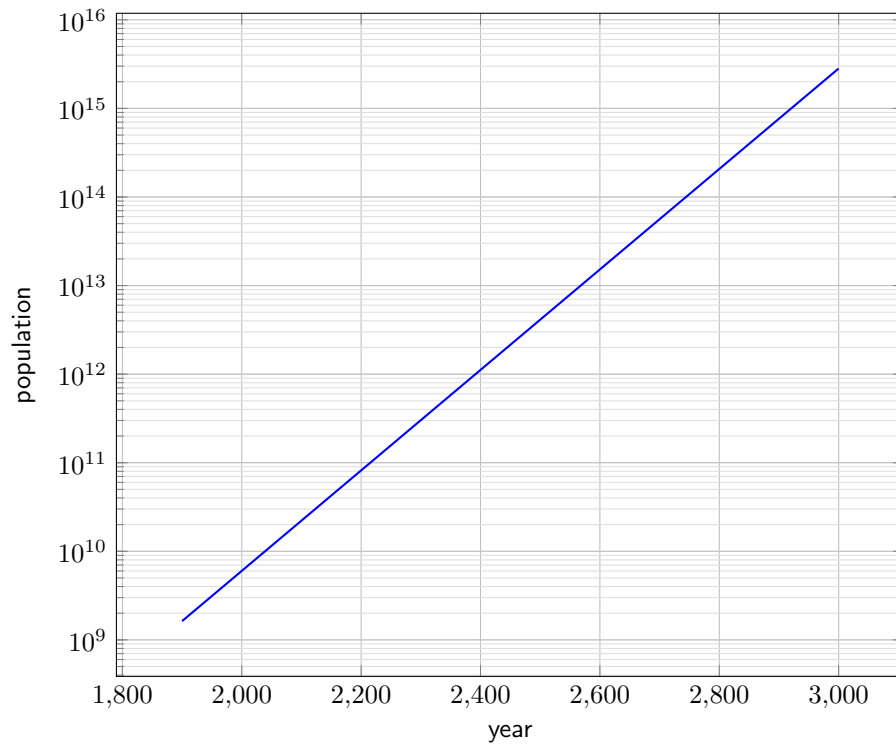
Below in red I've plotted the population across past years since 1900, well past the industrial revolution, my data comes from Our World In Data. In blue plotted is a function:

$$y = 1627883132 \times 1.01314982563501^{x-1900}$$

Which is simply exponential growth from the starting number of people (1627883132) in 1900 that both plots begin at, multiplied by 1.01314982563501 for every passing year to represent $\approx 1.315\%$ increase in population per year.



Let us consider the following millenium, and go all the way out to the year 3000 using this same roughly 1.315% growth per year:



Here are some precise numbers to be extracted from this:

year	exact population	rough population	order of magnitude
2000	6 011 607 607	6×10^9	billions
2050	11 552 456 090	11.5×10^9	tens of billions
2100	22 200 258 306	22.2×10^9	tens of billions
2150	42 662 050 824	42.7×10^9	tens of billions
2200	81 983 306 474	82×10^9	tens of billions
2250	157 546 634 788	157.5×10^9	hundreds of billions
2300	302 756 051 205	302.8×10^9	hundreds of billions
2350	581 803 772 987	581.8×10^9	hundreds of billions
2400	1 118 047 447 489	1.12×10^{12}	trillions
2450	2 148 542 434 542	2.15×10^{12}	trillions
2500	4 128 836 037 678	4.13×10^{12}	trillions
2600	15 247 373 503 074	15.2×10^{12}	tens of trillions
2700	56 307 006 773 998	56.3×10^{12}	tens of trillions
2800	207 936 075 758 080	207.9×10^{12}	hundreds of trillions
2900	767 886 877 297 784	767.9×10^{12}	hundreds of trillions
3000	2 835 728 500 580 920	2.8×10^{15}	quadrillions

year	population
2025	8.3336E+09
2030	8.8961E+09
2035	9.4966E+09
2040	1.0138E+10
2045	1.0822E+10
2050	1.1552E+10
2055	1.2332E+10
2060	1.3165E+10
2065	1.4053E+10
2070	1.5002E+10
2075	1.6015E+10
2080	1.7096E+10
2085	1.8250E+10
2090	1.9481E+10
2095	2.0796E+10
2100	2.2200E+10
2105	2.3699E+10
2110	2.5298E+10
2115	2.7006E+10
2120	2.8829E+10
2125	3.0775E+10
2130	3.2852E+10
2135	3.5070E+10
2140	3.7437E+10
2145	3.9964E+10
2150	4.2662E+10
2155	4.5542E+10
2160	4.8616E+10
2165	5.1898E+10
2170	5.5401E+10
2175	5.9140E+10
2180	6.3132E+10
2185	6.7394E+10
2190	7.1943E+10
2195	7.6799E+10
2200	8.1983E+10
2205	8.7517E+10
2210	9.3425E+10
2215	9.9731E+10
2220	1.0646E+11
2225	1.1365E+11
2230	1.2132E+11
2235	1.2951E+11
2240	1.3825E+11
2245	1.4758E+11
2250	1.5755E+11
2255	1.6818E+11
2260	1.7953E+11
2265	1.9165E+11

year	population
2270	2.0459E+11
2275	2.1840E+11
2280	2.3314E+11
2285	2.4888E+11
2290	2.6568E+11
2295	2.8361E+11
2300	3.0276E+11
2305	3.2319E+11
2310	3.4501E+11
2315	3.6830E+11
2320	3.9316E+11
2325	4.1970E+11
2330	4.4803E+11
2335	4.7827E+11
2340	5.1055E+11
2345	5.4501E+11
2350	5.8180E+11
2355	6.2108E+11
2360	6.6300E+11
2365	7.0775E+11
2370	7.5553E+11
2375	8.0653E+11
2380	8.6097E+11
2385	9.1908E+11
2390	9.8112E+11
2395	1.0474E+12
2400	1.1180E+12
2405	1.1935E+12
2410	1.2741E+12
2415	1.3601E+12
2420	1.4519E+12
2425	1.5499E+12
2430	1.6545E+12
2435	1.7662E+12
2440	1.8854E+12
2445	2.0127E+12
2450	2.1485E+12
2455	2.2936E+12
2460	2.4484E+12
2465	2.6137E+12
2470	2.7901E+12
2475	2.9784E+12
2480	3.1795E+12
2485	3.3941E+12
2490	3.6232E+12
2495	3.8678E+12
2500	4.1288E+12
2505	4.4075E+12
2510	4.7051E+12

year	population
2515	5.0226E+12
2520	5.3617E+12
2525	5.7236E+12
2530	6.1100E+12
2535	6.5224E+12
2540	6.9627E+12
2545	7.4326E+12
2550	7.9343E+12
2555	8.4699E+12
2560	9.0417E+12
2565	9.6520E+12
2570	1.0304E+13
2575	1.0999E+13
2580	1.1741E+13
2585	1.2534E+13
2590	1.3380E+13
2595	1.4283E+13
2600	1.5247E+13
2605	1.6277E+13
2610	1.7375E+13
2615	1.8548E+13
2620	1.9800E+13
2625	2.1137E+13
2630	2.2563E+13
2635	2.4086E+13
2640	2.5712E+13
2645	2.7448E+13
2650	2.9301E+13
2655	3.1279E+13
2660	3.3390E+13
2665	3.5644E+13
2670	3.8050E+13
2675	4.0618E+13
2680	4.3360E+13
2685	4.6287E+13
2690	4.9411E+13
2695	5.2747E+13
2700	5.6307E+13
2705	6.0108E+13
2710	6.4165E+13
2715	6.8496E+13
2720	7.3120E+13
2725	7.8056E+13
2730	8.3324E+13
2735	8.8949E+13
2740	9.4953E+13
2745	1.0136E+14
2750	1.0820E+14
2755	1.1551E+14

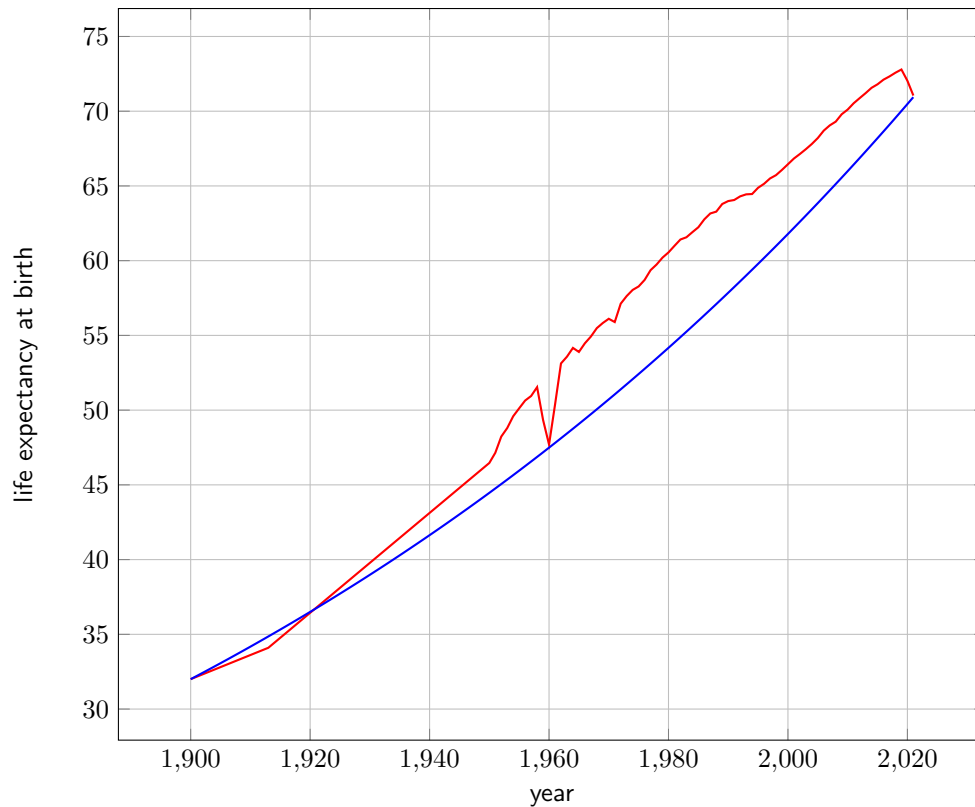
year	population
2760	1.2331E+14
2765	1.3163E+14
2770	1.4051E+14
2775	1.5000E+14
2780	1.6012E+14
2785	1.7093E+14
2790	1.8247E+14
2795	1.9479E+14
2800	2.0794E+14
2805	2.2197E+14
2810	2.3696E+14
2815	2.5295E+14
2820	2.7002E+14
2825	2.8825E+14
2830	3.0771E+14
2835	3.2848E+14
2840	3.5065E+14
2845	3.7432E+14
2850	3.9959E+14
2855	4.2656E+14
2860	4.5536E+14
2865	4.8609E+14
2870	5.1890E+14
2875	5.5393E+14
2880	5.9132E+14
2885	6.3124E+14
2890	6.7385E+14
2895	7.1933E+14
2900	7.6789E+14
2905	8.1972E+14
2910	8.7505E+14
2915	9.3412E+14
2920	9.9717E+14
2925	1.0645E+15
2930	1.1363E+15
2935	1.2130E+15
2940	1.2949E+15
2945	1.3823E+15
2950	1.4756E+15
2955	1.5752E+15
2960	1.6816E+15
2965	1.7951E+15
2970	1.9163E+15
2975	2.0456E+15
2980	2.1837E+15
2985	2.3311E+15
2990	2.4884E+15
2995	2.6564E+15
3000	2.8357E+15

2 exponential life expectancy at birth

Below in red I've plotted the global average life expectancy at birth between years 1900 and 2021 according to Our World In Data, and in blue an exponential function:

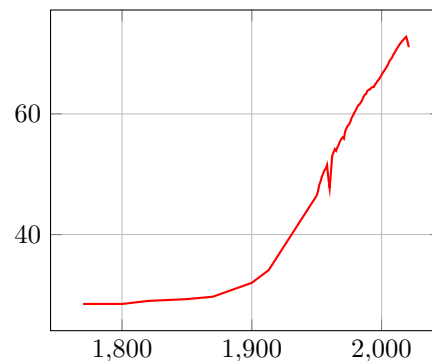
$$y = 32 \times 1.006613662^{x-1900}$$

Which represents a starting point at year 1900 with 32 years life expectancy at birth, and growth of roughly 0.66% per year.

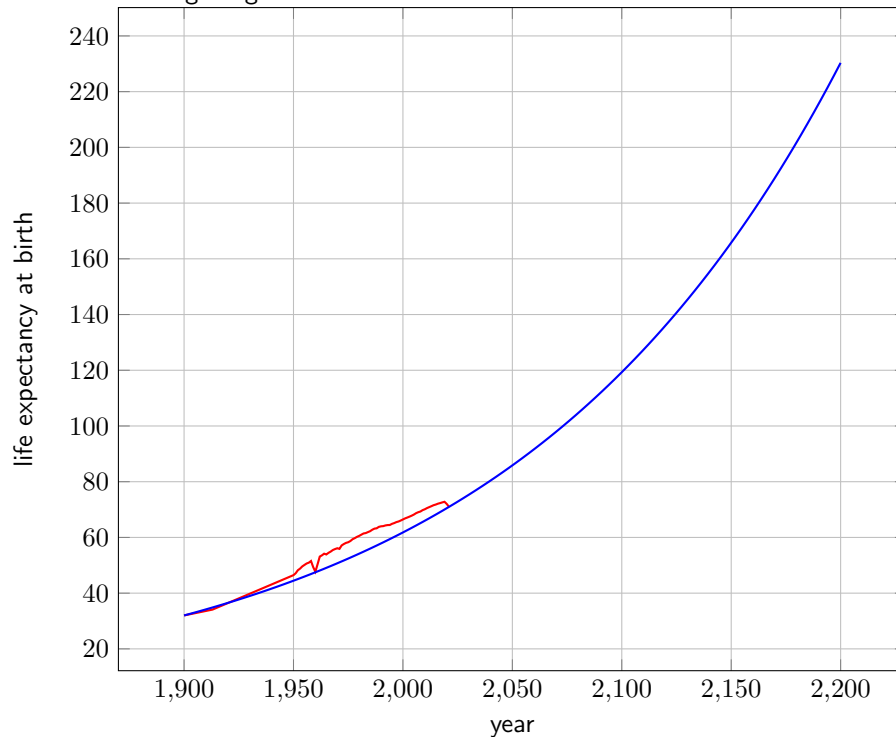


I should say, at least to my eye, the real data very much does not feel like it follows exponential growth, rather it looks more like it follows simple linear growth, but even more than that I would say the shape of the curve here is just very hard to guess because it doesn't even remotely look like any of the simple mathematical functions I know.

In the years before 1900 which I had not included above for industrial revolution reasons it becomes pretty much flat, showing little change as the years pass, making the actual curve look more like 2 straight lines that touch each other at an angle rather than any kind of gradual, let alone exponential growth.



Once again, let's extend this into the future, but for reasons you will see below I'm only going to extend this out to the year 2200 this time, and since the range is small enough for the real data to be visible I am including it again in red.



Here are some precise numbers to be extracted from this:

year	life expectancy at birth	increase over previous year
2020	70.58 y	0.4637 y
2030	75.39 y	0.4953 y
2040	80.53 y	0.5291 y
2050	86 y	0.5651 y
2060	91.88 y	0.6036 y
2070	98.14 y	0.6448 y
2080	104.82 y	0.6887 y
2090	111.97 y	0.7356 y
2100	119.59 y	0.7858 y
2110	127.74 y	0.8393 y
2120	136.45 y	0.8965 y
2130	145.75 y	0.9576 y
2137	152.63 y	1 y

It is a little pointless to continue past the year 2137 in this model as at this year, life expectancy begins to grow by more than 1 year every year, granting everyone born past that point theoretical immortality. The more widely known name for this effect is "longevity escape velocity" for those more interested.

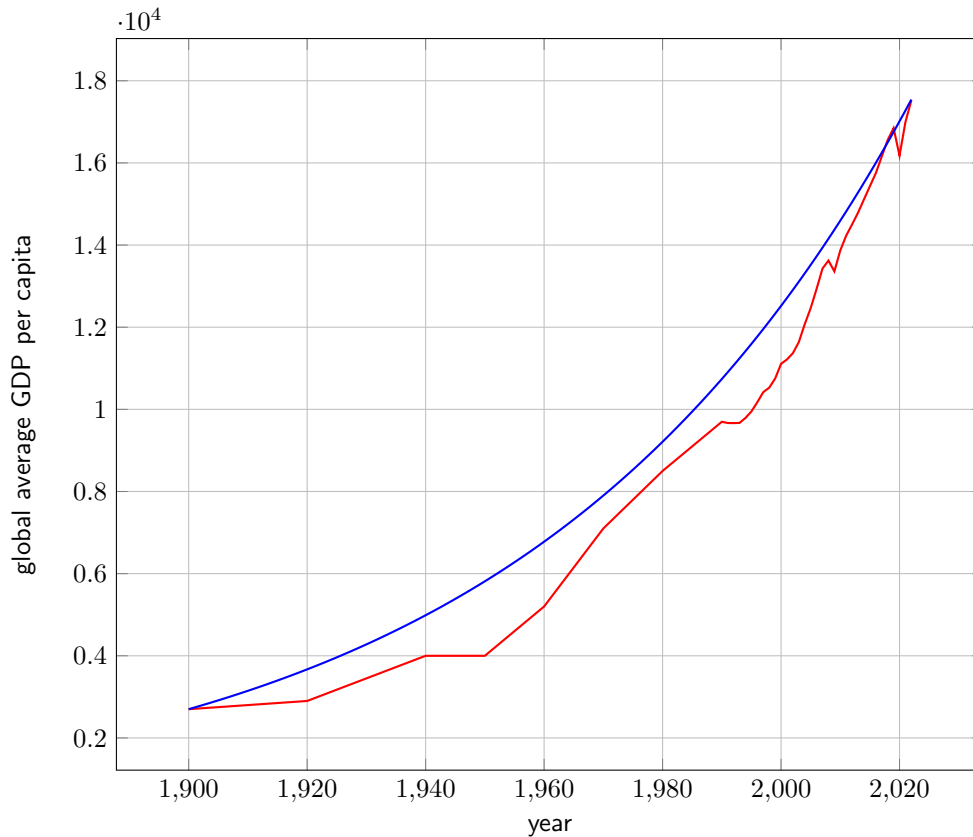
3 exponential GDP per capita growth

Going into this chapter it's important to remember that GDP per capita is not the same as the amount of money an average person earns, nor even really implies that money still exists in whatever society we are contemplating here - you might think of it as factoring in not just your personal earnings but costs to maintain the roads you drive on or the bus you take to work & such.

Below in red I've plotted the global average GDP per capita (expressed in international dollars at 2017 prices) between years 1900 and 2022 according to Our World In Data, and in blue an exponential function:

$$y = 2700 \times 1.015450028^{x-1900}$$

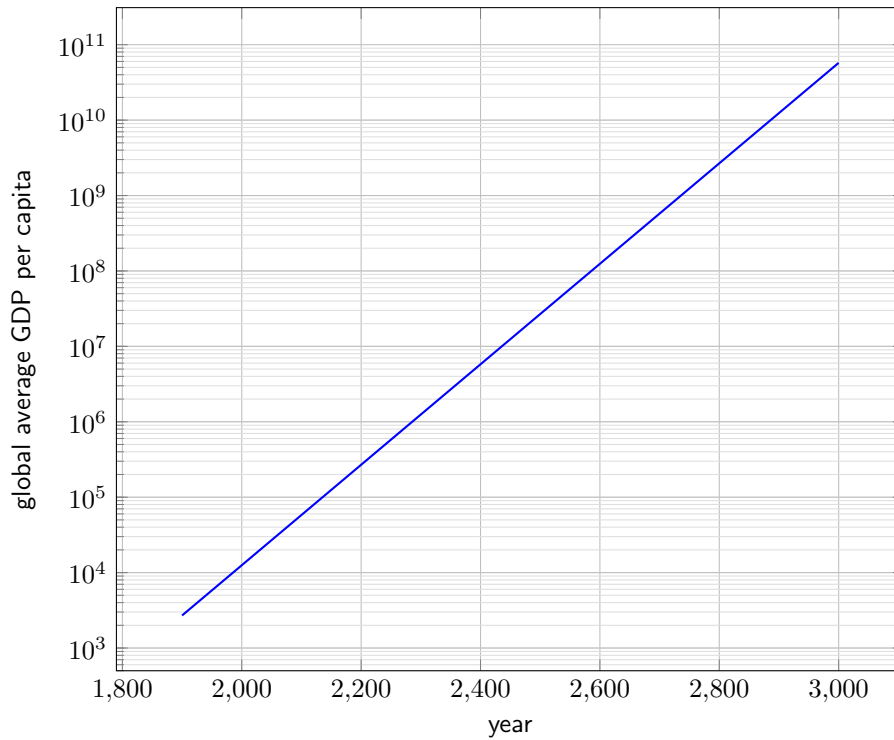
Which represents a starting point at year 1900 with 2700 international-\$, and growth of roughly 1.545% per year.



This appears to be a quicker rise than the population growth. There is also a third assumption I am forced to make given the two I've already made - if the number of people is growing exponentially, and the gross domestic product per person is also growing exponentially, then total global (universal?) GDP per capita must necessarily be also growing exponentially as just those 2 functions multiplied.

If I were to graph it here it would be some function $y = a \times 1.028803019^{(x-b)}$ or at least close to it, else the model is self-inconsistent.

Let us again consider this roughly 1.545% per year growth through the following millenium and go all the way to the year 3000:



Here are some precise numbers to be extracted from this:
(once again, in international dollars, at 2017 prices)

year	exact GDP p.c.	rough GDP p.c.	order of magnitude
2000	12 508.9 \$	12.5×10^3 \$	tens of thousands
2050	26 924.47 \$	26.9×10^3 \$	tens of thousands
2100	57 952.87 \$	58×10^3 \$	tens of thousands
2150	124 739.13 \$	124.7×10^3 \$	hundreds of thousands
2200	268 491.49 \$	268.5×10^3 \$	hundreds of thousands
2250	577 907.5 \$	577.9×10^3 \$	hundreds of thousands
2300	1 243 901.89 \$	1.2×10^6 \$	millions
2350	2 677 404.09 \$	2.7×10^6 \$	millions
2400	5 762 908.41 \$	5.8×10^6 \$	millions
2450	12 404 221.5 \$	12.4×10^6 \$	tens of millions
2500	26 699 142.17 \$	26.7×10^6 \$	tens of millions
2600	123 695 214.67 \$	123.7×10^6 \$	hundreds of millions
2700	573 071 075.94 \$	573×10^6 \$	hundreds of millions
2800	2 654 997 276.55 \$	2.7×10^9 \$	billions
2900	12 300 412 347.57 \$	12.3×10^9 \$	tens of billions
3000	56 986 929 989.15 \$	57×10^9 \$	tens of billions

year	GDP p.c.
2025	1.8352E+04
2030	1.9814E+04
2035	2.1393E+04
2040	2.3097E+04
2045	2.4938E+04
2050	2.6924E+04
2055	2.9070E+04
2060	3.1386E+04
2065	3.3886E+04
2070	3.6586E+04
2075	3.9501E+04
2080	4.2649E+04
2085	4.6046E+04
2090	4.9715E+04
2095	5.3676E+04
2100	5.7953E+04
2105	6.2570E+04
2110	6.7555E+04
2115	7.2938E+04
2120	7.8749E+04
2125	8.5023E+04
2130	9.1798E+04
2135	9.9112E+04
2140	1.0701E+05
2145	1.1553E+05
2150	1.2474E+05
2155	1.3468E+05
2160	1.4541E+05
2165	1.5699E+05
2170	1.6950E+05
2175	1.8301E+05
2180	1.9759E+05
2185	2.1333E+05
2190	2.3033E+05
2195	2.4868E+05
2200	2.6849E+05
2205	2.8988E+05
2210	3.1298E+05
2215	3.3792E+05
2220	3.6484E+05
2225	3.9391E+05
2230	4.2529E+05
2235	4.5918E+05
2240	4.9576E+05
2245	5.3526E+05
2250	5.7791E+05
2255	6.2395E+05
2260	6.7366E+05
2265	7.2734E+05

year	GDP p.c.
2270	7.8529E+05
2275	8.4786E+05
2280	9.1541E+05
2285	9.8834E+05
2290	1.0671E+06
2295	1.1521E+06
2300	1.2439E+06
2305	1.3430E+06
2310	1.4500E+06
2315	1.5655E+06
2320	1.6903E+06
2325	1.8249E+06
2330	1.9703E+06
2335	2.1273E+06
2340	2.2968E+06
2345	2.4798E+06
2350	2.6774E+06
2355	2.8907E+06
2360	3.1210E+06
2365	3.3697E+06
2370	3.6382E+06
2375	3.9281E+06
2380	4.2410E+06
2385	4.5789E+06
2390	4.9437E+06
2395	5.3376E+06
2400	5.7629E+06
2405	6.2221E+06
2410	6.7178E+06
2415	7.2530E+06
2420	7.8309E+06
2425	8.4548E+06
2430	9.1285E+06
2435	9.8558E+06
2440	1.0641E+07
2445	1.1489E+07
2450	1.2404E+07
2455	1.3393E+07
2460	1.4460E+07
2465	1.5612E+07
2470	1.6855E+07
2475	1.8198E+07
2480	1.9648E+07
2485	2.1214E+07
2490	2.2904E+07
2495	2.4729E+07
2500	2.6699E+07
2505	2.8826E+07
2510	3.1123E+07

year	GDP p.c.
2515	3.3603E+07
2520	3.6280E+07
2525	3.9171E+07
2530	4.2292E+07
2535	4.5661E+07
2540	4.9299E+07
2545	5.3227E+07
2550	5.7468E+07
2555	6.2047E+07
2560	6.6990E+07
2565	7.2327E+07
2570	7.8090E+07
2575	8.4312E+07
2580	9.1029E+07
2585	9.8282E+07
2590	1.0611E+08
2595	1.1457E+08
2600	1.2370E+08
2605	1.3355E+08
2610	1.4419E+08
2615	1.5568E+08
2620	1.6808E+08
2625	1.8147E+08
2630	1.9593E+08
2635	2.1154E+08
2640	2.2840E+08
2645	2.4660E+08
2650	2.6624E+08
2655	2.8746E+08
2660	3.1036E+08
2665	3.3509E+08
2670	3.6179E+08
2675	3.9061E+08
2680	4.2173E+08
2685	4.5533E+08
2690	4.9161E+08
2695	5.3078E+08
2700	5.7307E+08
2705	6.1873E+08
2710	6.6803E+08
2715	7.2125E+08
2720	7.7872E+08
2725	8.4076E+08
2730	9.0775E+08
2735	9.8007E+08
2740	1.0582E+09
2745	1.1425E+09
2750	1.2335E+09
2755	1.3318E+09

year	GDP p.c.
2760	1.4379E+09
2765	1.5524E+09
2770	1.6761E+09
2775	1.8097E+09
2780	1.9539E+09
2785	2.1095E+09
2790	2.2776E+09
2795	2.4591E+09
2800	2.6550E+09
2805	2.8665E+09
2810	3.0949E+09
2815	3.3415E+09
2820	3.6077E+09
2825	3.8952E+09
2830	4.2055E+09
2835	4.5406E+09
2840	4.9024E+09
2845	5.2930E+09
2850	5.7147E+09
2855	6.1700E+09
2860	6.6616E+09
2865	7.1923E+09
2870	7.7654E+09
2875	8.3841E+09
2880	9.0521E+09
2885	9.7733E+09
2890	1.0552E+10
2895	1.1393E+10
2900	1.2300E+10
2905	1.3280E+10
2910	1.4339E+10
2915	1.5481E+10
2920	1.6714E+10
2925	1.8046E+10
2930	1.9484E+10
2935	2.1036E+10
2940	2.2712E+10
2945	2.4522E+10
2950	2.6476E+10
2955	2.8585E+10
2960	3.0863E+10
2965	3.3322E+10
2970	3.5976E+10
2975	3.8843E+10
2980	4.1938E+10
2985	4.5279E+10
2990	4.8887E+10
2995	5.2782E+10
3000	5.6987E+10

4 exponential global total GDP

As mentioned in the previous section, given the population growth and average GDP per capita growth rates, the total GDP growth rate is expected to be 2.88% per year.

Below in red I've plotted the global total GDP (expressed in international dollars at 2017 prices) between years 1900 and 2022 according to Our World In Data, and in blue an exponential function:

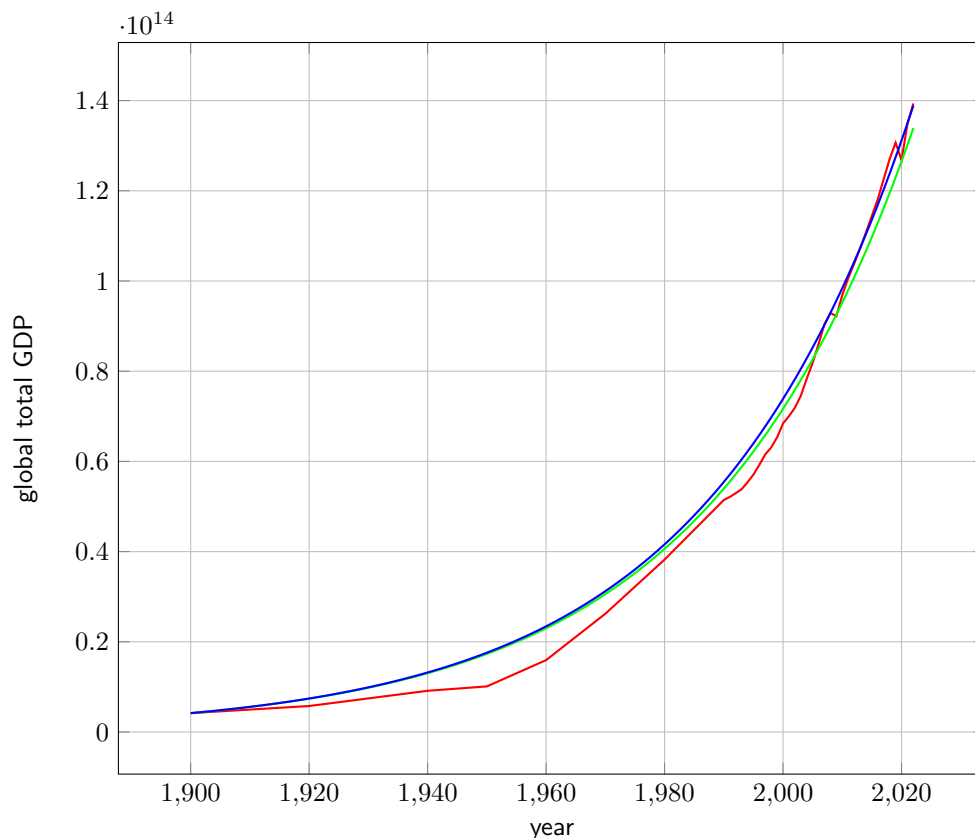
$$y = 4188242000000 \times 1.029144185^{x-1900}$$

Which represents a starting point at year 1900 with 4188242000000 international-\$, and growth of roughly 2.91% per year.

Additionally I've plotted in green the function mentioned previously which I deduced from the population and GDP per capita growth:

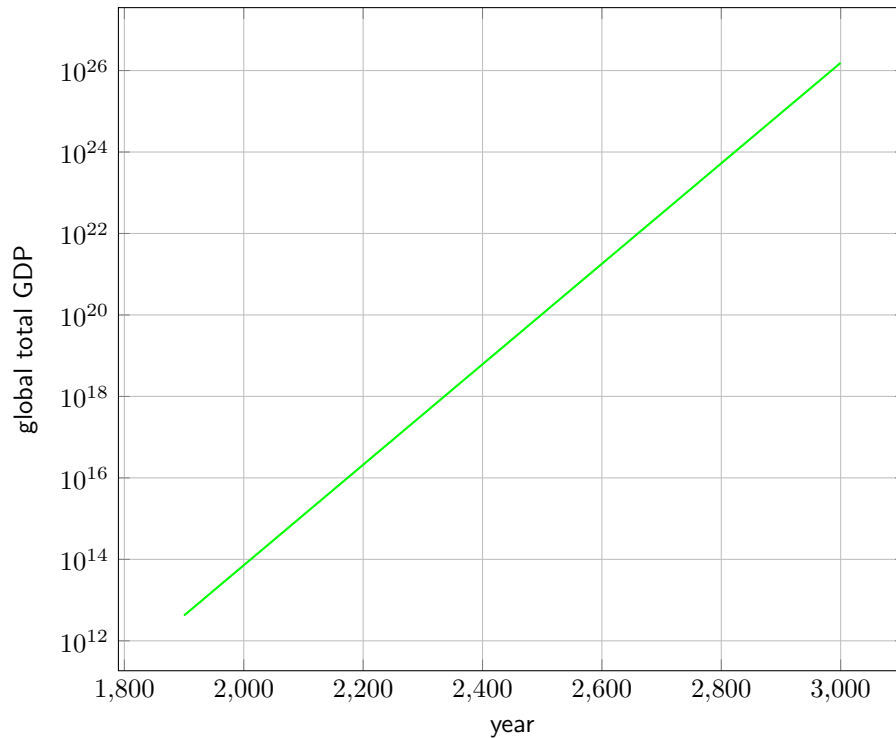
$$y = 4188242000000 \times 1.028803019^{x-1900}$$

Which represents growth at 2.88% per year from the same starting point.



As can be seen, the data is consistent with previous predictions. The blue curve intersects the final point of the data here (year 2022) as it should given how it is calculated, while the green deduced from population and gdp per capita data intersects the point for 2021 year, possibly reflecting the fact my data for global average gdp per capita runs from 1900 to 2021 and not 2022.

Let's extrapolate the found rates of growth - both 2.88% and 2.91% - all the way to the year 3000. First, **2.88%** which is the rate consistent with GDP per capita & population growth shown earlier:



Here are the numbers that can be extracted from this:

year	exact GDP (exp. 2.88%)	rough
2000	71 656 364 205 474 \$	7.17×10^{13} \$
2050	296 391 857 890 027 \$	2.96×10^{14} \$
2100	1 225 964 147 044 890 \$	1.23×10^{15} \$
2150	5 070 949 318 712 950 \$	5.07×10^{15} \$
2200	20 974 942 093 485 100 \$	2.10×10^{16} \$
2250	86 758 547 201 713 000 \$	8.68×10^{16} \$
2300	358 858 941 254 946 000 \$	3.59×10^{17} \$
2350	1 484 346 428 936 960 000 \$	1.48×10^{18} \$
2400	6 139 694 648 245 090 000 \$	6.14×10^{18} \$
2450	25 395 588 010 196 600 000 \$	2.54×10^{19} \$
2500	105 043 642 613 071 000 000 \$	1.05×10^{20} \$
2600	1 797 184 955 537 870 000 000 \$	1.80×10^{21} \$
2700	30 747 922 330 806 300 000 000 \$	3.07×10^{22} \$
2800	526 064 234 372 774 000 000 000 \$	5.26×10^{23} \$
2900	9 000 399 302 067 420 000 000 000 \$	9.00×10^{24} \$
3000	153 987 255 364 814 000 000 000 000 \$	1.54×10^{26} \$

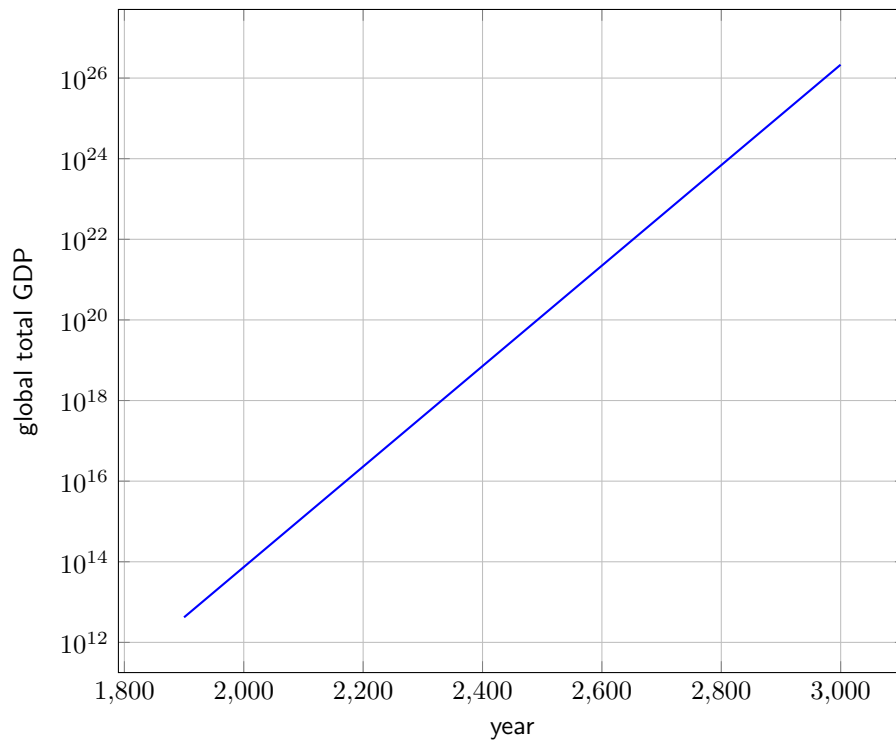
year	GDP (2.88%)
2025	1.4573E+14
2030	1.6797E+14
2035	1.9359E+14
2040	2.2312E+14
2045	2.5716E+14
2050	2.9639E+14
2055	3.4161E+14
2060	3.9372E+14
2065	4.5378E+14
2070	5.2301E+14
2075	6.0280E+14
2080	6.9476E+14
2085	8.0075E+14
2090	9.2290E+14
2095	1.0637E+15
2100	1.2260E+15
2105	1.4130E+15
2110	1.6285E+15
2115	1.8770E+15
2120	2.1633E+15
2125	2.4934E+15
2130	2.8737E+15
2135	3.3121E+15
2140	3.8174E+15
2145	4.3998E+15
2150	5.0709E+15
2155	5.8445E+15
2160	6.7361E+15
2165	7.7638E+15
2170	8.9482E+15
2175	1.0313E+16
2180	1.1887E+16
2185	1.3700E+16
2190	1.5790E+16
2195	1.8199E+16
2200	2.0975E+16
2205	2.4175E+16
2210	2.7863E+16
2215	3.2113E+16
2220	3.7012E+16
2225	4.2659E+16
2230	4.9166E+16
2235	5.6667E+16
2240	6.5312E+16
2245	7.5275E+16
2250	8.6759E+16
2255	9.9994E+16
2260	1.1525E+17
2265	1.3283E+17

year	GDP (2.88%)
2270	1.5309E+17
2275	1.7645E+17
2280	2.0337E+17
2285	2.3439E+17
2290	2.7015E+17
2295	3.1136E+17
2300	3.5886E+17
2305	4.1360E+17
2310	4.7670E+17
2315	5.4942E+17
2320	6.3324E+17
2325	7.2984E+17
2330	8.4118E+17
2335	9.6951E+17
2340	1.1174E+18
2345	1.2879E+18
2350	1.4843E+18
2355	1.7108E+18
2360	1.9718E+18
2365	2.2726E+18
2370	2.6193E+18
2375	3.0188E+18
2380	3.4794E+18
2385	4.0102E+18
2390	4.6219E+18
2395	5.3270E+18
2400	6.1397E+18
2405	7.0763E+18
2410	8.1558E+18
2415	9.4001E+18
2420	1.0834E+19
2425	1.2487E+19
2430	1.4392E+19
2435	1.6587E+19
2440	1.9118E+19
2445	2.2034E+19
2450	2.5396E+19
2455	2.9270E+19
2460	3.3735E+19
2465	3.8881E+19
2470	4.4813E+19
2475	5.1649E+19
2480	5.9529E+19
2485	6.8610E+19
2490	7.9077E+19
2495	9.1140E+19
2500	1.0504E+20
2505	1.2107E+20
2510	1.3954E+20

year	GDP (2.88%)
2515	1.6082E+20
2520	1.8536E+20
2525	2.1364E+20
2530	2.4623E+20
2535	2.8379E+20
2540	3.2708E+20
2545	3.7698E+20
2550	4.3449E+20
2555	5.0077E+20
2560	5.7717E+20
2565	6.6522E+20
2570	7.6670E+20
2575	8.8366E+20
2580	1.0185E+21
2585	1.1738E+21
2590	1.3529E+21
2595	1.5593E+21
2600	1.7972E+21
2605	2.0714E+21
2610	2.3873E+21
2615	2.7515E+21
2620	3.1713E+21
2625	3.6551E+21
2630	4.2127E+21
2635	4.8554E+21
2640	5.5961E+21
2645	6.4498E+21
2650	7.4337E+21
2655	8.5677E+21
2660	9.8748E+21
2665	1.1381E+22
2670	1.3117E+22
2675	1.5119E+22
2680	1.7425E+22
2685	2.0083E+22
2690	2.3147E+22
2695	2.6678E+22
2700	3.0748E+22
2705	3.5439E+22
2710	4.0845E+22
2715	4.7076E+22
2720	5.4258E+22
2725	6.2535E+22
2730	7.2075E+22
2735	8.3070E+22
2740	9.5743E+22
2745	1.1035E+23
2750	1.2718E+23
2755	1.4658E+23

year	GDP (2.88%)
2760	1.6895E+23
2765	1.9472E+23
2770	2.2443E+23
2775	2.5866E+23
2780	2.9812E+23
2785	3.4360E+23
2790	3.9602E+23
2795	4.5643E+23
2800	5.2606E+23
2805	6.0632E+23
2810	6.9881E+23
2815	8.0542E+23
2820	9.2829E+23
2825	1.0699E+24
2830	1.2331E+24
2835	1.4212E+24
2840	1.6381E+24
2845	1.8879E+24
2850	2.1760E+24
2855	2.5079E+24
2860	2.8905E+24
2865	3.3315E+24
2870	3.8397E+24
2875	4.4254E+24
2880	5.1006E+24
2885	5.8787E+24
2890	6.7755E+24
2895	7.8091E+24
2900	9.0004E+24
2905	1.0373E+25
2910	1.1956E+25
2915	1.3780E+25
2920	1.5882E+25
2925	1.8305E+25
2930	2.1097E+25
2935	2.4316E+25
2940	2.8025E+25
2945	3.2301E+25
2950	3.7228E+25
2955	4.2908E+25
2960	4.9453E+25
2965	5.6998E+25
2970	6.5693E+25
2975	7.5714E+25
2980	8.7265E+25
2985	1.0058E+26
2990	1.1592E+26
2995	1.3361E+26
3000	1.5399E+26

Now again, for **2.91%** which is the rate deduced from the actual data for global total gdp gathered from Our World In Data:



Here are the numbers that can be extracted from this:

year	exact GDP (exp. 2.91%)	rough
2000	74 072 024 713 093 \$	7.41×10^{13} \$
2050	311 505 299 969 375 \$	3.12×10^{14} \$
2100	1 310 016 194 168 611 \$	1.31×10^{15} \$
2150	5 509 191 750 999 848 \$	5.51×10^{15} \$
2200	23 168 563 781 417 100 \$	2.32×10^{16} \$
2250	97 433 956 187 161 900 \$	9.74×10^{16} \$
2300	409 752 451 979 616 000 \$	4.10×10^{17} \$
2350	1 723 188 490 681 750 000 \$	1.72×10^{18} \$
2400	7 246 762 185 490 900 000 \$	7.25×10^{18} \$
2450	30 475 808 338 462 100 000 \$	3.05×10^{19} \$
2500	128 164 119 383 167 000 000 \$	1.28×10^{20} \$
2600	2 266 673 181 320 900 000 000 \$	2.27×10^{21} \$
2700	40 087 719 836 463 000 000 000 \$	4.01×10^{22} \$
2800	708 979 704 233 437 000 000 000 \$	7.09×10^{23} \$
2900	12 538 807 970 757 400 000 000 000 \$	1.25×10^{25} \$
3000	221 757 695 444 216 000 000 000 000 \$	2.22×10^{26} \$

year	GDP (2.91%)
2025	1.5190E+14
2030	1.7536E+14
2035	2.0245E+14
2040	2.3372E+14
2045	2.6983E+14
2050	3.1151E+14
2055	3.5962E+14
2060	4.1517E+14
2065	4.7930E+14
2070	5.5334E+14
2075	6.3881E+14
2080	7.3748E+14
2085	8.5140E+14
2090	9.8291E+14
2095	1.1347E+15
2100	1.3100E+15
2105	1.5124E+15
2110	1.7460E+15
2115	2.0157E+15
2120	2.3270E+15
2125	2.6865E+15
2130	3.1014E+15
2135	3.5805E+15
2140	4.1336E+15
2145	4.7721E+15
2150	5.5092E+15
2155	6.3602E+15
2160	7.3426E+15
2165	8.4768E+15
2170	9.7862E+15
2175	1.1298E+16
2180	1.3043E+16
2185	1.5058E+16
2190	1.7383E+16
2195	2.0069E+16
2200	2.3169E+16
2205	2.6747E+16
2210	3.0879E+16
2215	3.5649E+16
2220	4.1155E+16
2225	4.7512E+16
2230	5.4851E+16
2235	6.3324E+16
2240	7.3105E+16
2245	8.4397E+16
2250	9.7434E+16
2255	1.1248E+17
2260	1.2986E+17
2265	1.4992E+17

year	GDP (2.91%)
2270	1.7308E+17
2275	1.9981E+17
2280	2.3067E+17
2285	2.6630E+17
2290	3.0744E+17
2295	3.5493E+17
2300	4.0975E+17
2305	4.7305E+17
2310	5.4611E+17
2315	6.3047E+17
2320	7.2786E+17
2325	8.4029E+17
2330	9.7008E+17
2335	1.1199E+18
2340	1.2929E+18
2345	1.4926E+18
2350	1.7232E+18
2355	1.9894E+18
2360	2.2967E+18
2365	2.6514E+18
2370	3.0610E+18
2375	3.5338E+18
2380	4.0796E+18
2385	4.7098E+18
2390	5.4373E+18
2395	6.2772E+18
2400	7.2468E+18
2405	8.3661E+18
2410	9.6584E+18
2415	1.1150E+19
2420	1.2873E+19
2425	1.4861E+19
2430	1.7157E+19
2435	1.9807E+19
2440	2.2866E+19
2445	2.6398E+19
2450	3.0476E+19
2455	3.5183E+19
2460	4.0618E+19
2465	4.6892E+19
2470	5.4135E+19
2475	6.2497E+19
2480	7.2151E+19
2485	8.3296E+19
2490	9.6162E+19
2495	1.1102E+20
2500	1.2816E+20
2505	1.4796E+20
2510	1.7082E+20

year	GDP (2.91%)
2515	1.9720E+20
2520	2.2766E+20
2525	2.6283E+20
2530	3.0343E+20
2535	3.5030E+20
2540	4.0440E+20
2545	4.6687E+20
2550	5.3899E+20
2555	6.2224E+20
2560	7.1836E+20
2565	8.2932E+20
2570	9.5742E+20
2575	1.1053E+21
2580	1.2760E+21
2585	1.4731E+21
2590	1.7007E+21
2595	1.9634E+21
2600	2.2667E+21
2605	2.6168E+21
2610	3.0210E+21
2615	3.4876E+21
2620	4.0264E+21
2625	4.6483E+21
2630	5.3663E+21
2635	6.1952E+21
2640	7.1522E+21
2645	8.2569E+21
2650	9.5324E+21
2655	1.1005E+22
2660	1.2705E+22
2665	1.4667E+22
2670	1.6933E+22
2675	1.9548E+22
2680	2.2568E+22
2685	2.6054E+22
2690	3.0078E+22
2695	3.4724E+22
2700	4.0088E+22
2705	4.6280E+22
2710	5.3429E+22
2715	6.1681E+22
2720	7.1209E+22
2725	8.2209E+22
2730	9.4907E+22
2735	1.0957E+23
2740	1.2649E+23
2745	1.4603E+23
2750	1.6859E+23
2755	1.9463E+23

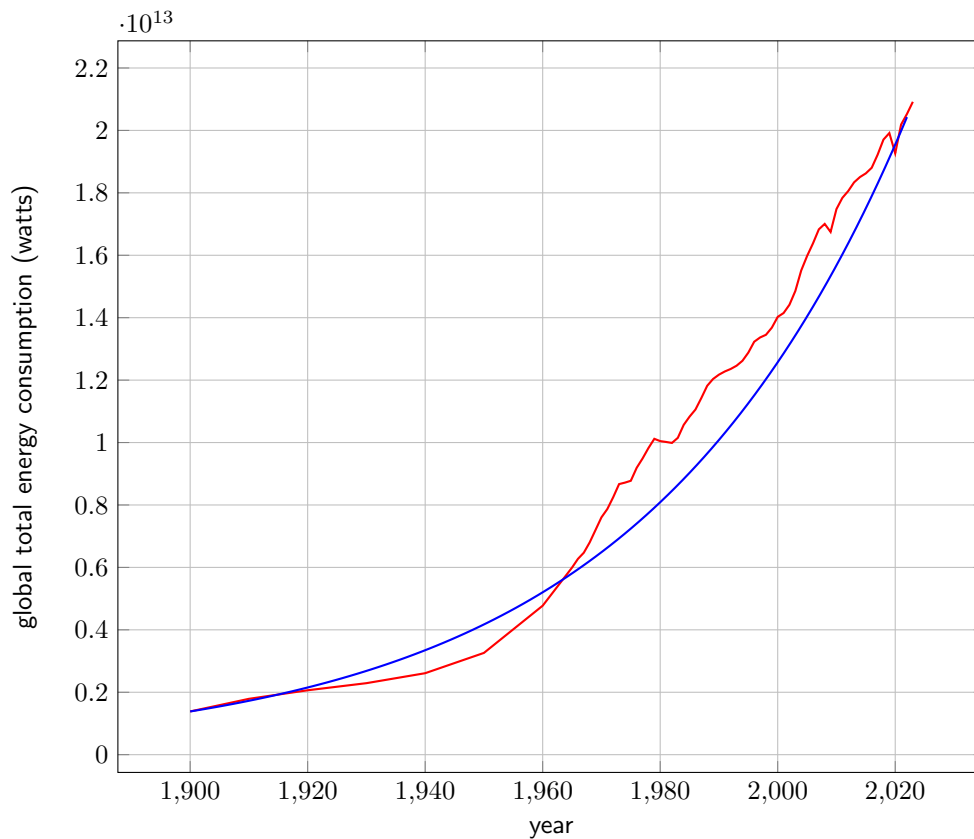
year	GDP (2.91%)
2760	2.2469E+23
2765	2.5940E+23
2770	2.9947E+23
2775	3.4572E+23
2780	3.9913E+23
2785	4.6078E+23
2790	5.3195E+23
2795	6.1412E+23
2800	7.0898E+23
2805	8.1849E+23
2810	9.4492E+23
2815	1.0909E+24
2820	1.2594E+24
2825	1.4539E+24
2830	1.6785E+24
2835	1.9378E+24
2840	2.2371E+24
2845	2.5826E+24
2850	2.9816E+24
2855	3.4421E+24
2860	3.9738E+24
2865	4.5876E+24
2870	5.2963E+24
2875	6.1144E+24
2880	7.0588E+24
2885	8.1492E+24
2890	9.4079E+24
2895	1.0861E+25
2900	1.2539E+25
2905	1.4476E+25
2910	1.6712E+25
2915	1.9293E+25
2920	2.2273E+25
2925	2.5714E+25
2930	2.9685E+25
2935	3.4271E+25
2940	3.9564E+25
2945	4.5676E+25
2950	5.2731E+25
2955	6.0876E+25
2960	7.0280E+25
2965	8.1135E+25
2970	9.3668E+25
2975	1.0814E+26
2980	1.2484E+26
2985	1.4412E+26
2990	1.6639E+26
2995	1.9209E+26
3000	2.2176E+26

5 exponential total energy consumption

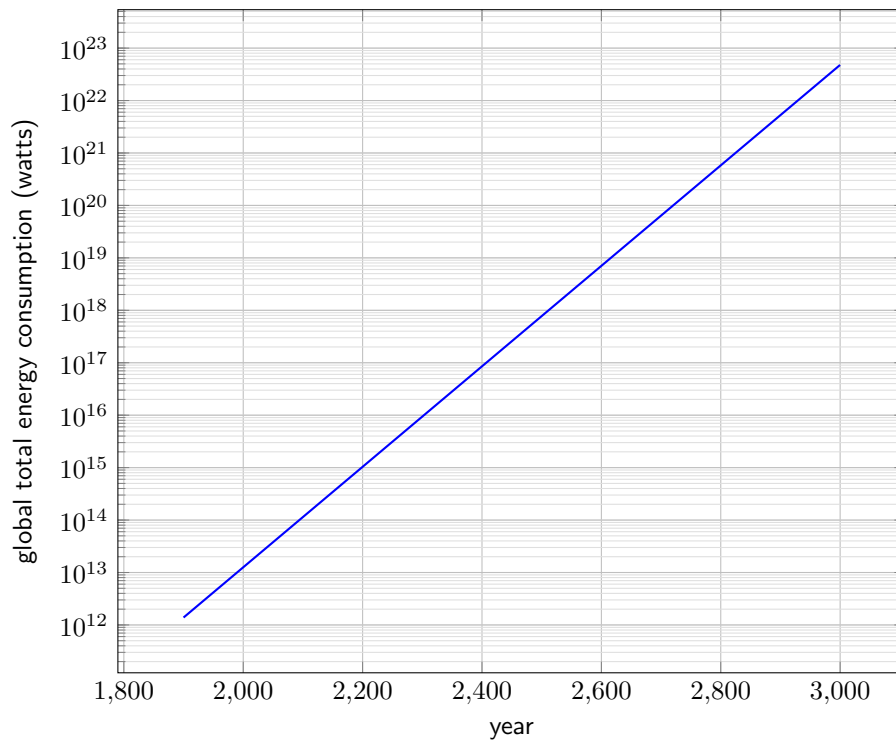
Below in red I've plotted the global energy consumption between years 1900 and 2023 according to Our World In Data (converted from terawatt-hours per year to watts), and in blue an exponential function:

$$y = 1384842719178.08 \times 1.022318254^{x-1900}$$

Which represents a starting point at year 1900 with 1384842719178 watts, and growth of roughly 2.23% per year.



Let's again stretch this out to the year 3000:



Here are some exact numbers that can be extracted from this:

year	exact	rough	kardashev level
2000	12 589 601 767 577 W	1.26E+13 W	0.71
2050	37 959 262 246 954 W	3.80E+13 W	0.76
2100	114 452 038 828 097 W	1.14E+14 W	0.81
2150	345 087 560 097 656 W	3.45E+14 W	0.85
2200	1 040 483 204 611 280 W	1.04E+15 W	0.90
2250	3 137 190 163 481 378 W	3.14E+15 W	0.95
2300	9 459 030 264 233 090 W	9.46E+15 W	1.00
2350	28 520 188 090 985 200 W	2.85E+16 W	1.05
2400	85 992 020 960 208 200 W	8.60E+16 W	1.09
2450	259 276 960 068 794 000 W	2.59E+17 W	1.14
2500	781 753 251 893 250 000 W	7.82E+17 W	1.19
2600	7 106 916 897 888 330 000 W	7.11E+18 W	1.29
2700	64 608 964 108 776 900 000 W	6.46E+19 W	1.38
2800	587 359 934 439 298 000 000 W	5.87E+20 W	1.48
2900	5 339 687 725 122 810 000 000 W	5.34E+21 W	1.57
3000	48 543 088 028 374 600 000 000 W	4.85E+22 W	1.67

year	Watts
2025	2.1861E+13
2030	2.4412E+13
2035	2.7260E+13
2040	3.0441E+13
2045	3.3993E+13
2050	3.7959E+13
2055	4.2389E+13
2060	4.7335E+13
2065	5.2858E+13
2070	5.9026E+13
2075	6.5913E+13
2080	7.3604E+13
2085	8.2192E+13
2090	9.1783E+13
2095	1.0249E+14
2100	1.1445E+14
2105	1.2781E+14
2110	1.4272E+14
2115	1.5937E+14
2120	1.7797E+14
2125	1.9874E+14
2130	2.2193E+14
2135	2.4782E+14
2140	2.7674E+14
2145	3.0903E+14
2150	3.4509E+14
2155	3.8535E+14
2160	4.3032E+14
2165	4.8053E+14
2170	5.3660E+14
2175	5.9921E+14
2180	6.6913E+14
2185	7.4721E+14
2190	8.3440E+14
2195	9.3176E+14
2200	1.0405E+15
2205	1.1619E+15
2210	1.2975E+15
2215	1.4489E+15
2220	1.6179E+15
2225	1.8067E+15
2230	2.0175E+15
2235	2.2529E+15
2240	2.5158E+15
2245	2.8094E+15
2250	3.1372E+15
2255	3.5033E+15
2260	3.9120E+15
2265	4.3685E+15

year	Watts
2270	4.8782E+15
2275	5.4475E+15
2280	6.0831E+15
2285	6.7929E+15
2290	7.5855E+15
2295	8.4706E+15
2300	9.4590E+15
2305	1.0563E+16
2310	1.1795E+16
2315	1.3172E+16
2320	1.4709E+16
2325	1.6425E+16
2330	1.8341E+16
2335	2.0481E+16
2340	2.2871E+16
2345	2.5540E+16
2350	2.8520E+16
2355	3.1848E+16
2360	3.5564E+16
2365	3.9714E+16
2370	4.4348E+16
2375	4.9523E+16
2380	5.5301E+16
2385	6.1754E+16
2390	6.8960E+16
2395	7.7007E+16
2400	8.5992E+16
2405	9.6026E+16
2410	1.0723E+17
2415	1.1974E+17
2420	1.3372E+17
2425	1.4932E+17
2430	1.6674E+17
2435	1.8620E+17
2440	2.0792E+17
2445	2.3218E+17
2450	2.5928E+17
2455	2.8953E+17
2460	3.2331E+17
2465	3.6104E+17
2470	4.0317E+17
2475	4.5021E+17
2480	5.0274E+17
2485	5.6141E+17
2490	6.2691E+17
2495	7.0007E+17
2500	7.8175E+17
2505	8.7297E+17
2510	9.7483E+17

year	Watts
2515	1.0886E+18
2520	1.2156E+18
2525	1.3574E+18
2530	1.5158E+18
2535	1.6927E+18
2540	1.8902E+18
2545	2.1108E+18
2550	2.3571E+18
2555	2.6321E+18
2560	2.9393E+18
2565	3.2822E+18
2570	3.6652E+18
2575	4.0929E+18
2580	4.5704E+18
2585	5.1038E+18
2590	5.6993E+18
2595	6.3643E+18
2600	7.1069E+18
2605	7.9362E+18
2610	8.8622E+18
2615	9.8963E+18
2620	1.1051E+19
2625	1.2341E+19
2630	1.3780E+19
2635	1.5388E+19
2640	1.7184E+19
2645	1.9189E+19
2650	2.1428E+19
2655	2.3929E+19
2660	2.6721E+19
2665	2.9839E+19
2670	3.3320E+19
2675	3.7208E+19
2680	4.1550E+19
2685	4.6398E+19
2690	5.1812E+19
2695	5.7858E+19
2700	6.4609E+19
2705	7.2148E+19
2710	8.0566E+19
2715	8.9967E+19
2720	1.0047E+20
2725	1.1219E+20
2730	1.2528E+20
2735	1.3990E+20
2740	1.5622E+20
2745	1.7445E+20
2750	1.9480E+20
2755	2.1753E+20

year	Watts
2760	2.4292E+20
2765	2.7126E+20
2770	3.0292E+20
2775	3.3826E+20
2780	3.7773E+20
2785	4.2181E+20
2790	4.7102E+20
2795	5.2599E+20
2800	5.8736E+20
2805	6.5590E+20
2810	7.3243E+20
2815	8.1789E+20
2820	9.1333E+20
2825	1.0199E+21
2830	1.1389E+21
2835	1.2718E+21
2840	1.4202E+21
2845	1.5859E+21
2850	1.7710E+21
2855	1.9776E+21
2860	2.2084E+21
2865	2.4660E+21
2870	2.7538E+21
2875	3.0751E+21
2880	3.4339E+21
2885	3.8346E+21
2890	4.2821E+21
2895	4.7817E+21
2900	5.3397E+21
2905	5.9627E+21
2910	6.6585E+21
2915	7.4355E+21
2920	8.3031E+21
2925	9.2719E+21
2930	1.0354E+22
2935	1.1562E+22
2940	1.2911E+22
2945	1.4418E+22
2950	1.6100E+22
2955	1.7978E+22
2960	2.0076E+22
2965	2.2419E+22
2970	2.5035E+22
2975	2.7956E+22
2980	3.1218E+22
2985	3.4861E+22
2990	3.8928E+22
2995	4.3471E+22
3000	4.8543E+22

year	Kardashev
2025	0.734
2030	0.739
2035	0.744
2040	0.748
2045	0.753
2050	0.758
2055	0.763
2060	0.768
2065	0.772
2070	0.777
2075	0.782
2080	0.787
2085	0.791
2090	0.796
2095	0.801
2100	0.806
2105	0.811
2110	0.815
2115	0.820
2120	0.825
2125	0.830
2130	0.835
2135	0.839
2140	0.844
2145	0.849
2150	0.854
2155	0.859
2160	0.863
2165	0.868
2170	0.873
2175	0.878
2180	0.883
2185	0.887
2190	0.892
2195	0.897
2200	0.902
2205	0.907
2210	0.911
2215	0.916
2220	0.921
2225	0.926
2230	0.930
2235	0.935
2240	0.940
2245	0.945
2250	0.950
2255	0.954
2260	0.959
2265	0.964

year	Kardashev
2270	0.969
2275	0.974
2280	0.978
2285	0.983
2290	0.988
2295	0.993
2300	0.998
2305	1.002
2310	1.007
2315	1.012
2320	1.017
2325	1.022
2330	1.026
2335	1.031
2340	1.036
2345	1.041
2350	1.046
2355	1.050
2360	1.055
2365	1.060
2370	1.065
2375	1.069
2380	1.074
2385	1.079
2390	1.084
2395	1.089
2400	1.093
2405	1.098
2410	1.103
2415	1.108
2420	1.113
2425	1.117
2430	1.122
2435	1.127
2440	1.132
2445	1.137
2450	1.141
2455	1.146
2460	1.151
2465	1.156
2470	1.161
2475	1.165
2480	1.170
2485	1.175
2490	1.180
2495	1.185
2500	1.189
2505	1.194
2510	1.199

year	Kardashev
2515	1.204
2520	1.208
2525	1.213
2530	1.218
2535	1.223
2540	1.228
2545	1.232
2550	1.237
2555	1.242
2560	1.247
2565	1.252
2570	1.256
2575	1.261
2580	1.266
2585	1.271
2590	1.276
2595	1.280
2600	1.285
2605	1.290
2610	1.295
2615	1.300
2620	1.304
2625	1.309
2630	1.314
2635	1.319
2640	1.324
2645	1.328
2650	1.333
2655	1.338
2660	1.343
2665	1.347
2670	1.352
2675	1.357
2680	1.362
2685	1.367
2690	1.371
2695	1.376
2700	1.381
2705	1.386
2710	1.391
2715	1.395
2720	1.400
2725	1.405
2730	1.410
2735	1.415
2740	1.419
2745	1.424
2750	1.429
2755	1.434

year	Kardashev
2760	1.439
2765	1.443
2770	1.448
2775	1.453
2780	1.458
2785	1.463
2790	1.467
2795	1.472
2800	1.477
2805	1.482
2810	1.486
2815	1.491
2820	1.496
2825	1.501
2830	1.506
2835	1.510
2840	1.515
2845	1.520
2850	1.525
2855	1.530
2860	1.534
2865	1.539
2870	1.544
2875	1.549
2880	1.554
2885	1.558
2890	1.563
2895	1.568
2900	1.573
2905	1.578
2910	1.582
2915	1.587
2920	1.592
2925	1.597
2930	1.602
2935	1.606
2940	1.611
2945	1.616
2950	1.621
2955	1.625
2960	1.630
2965	1.635
2970	1.640
2975	1.645
2980	1.649
2985	1.654
2990	1.659
2995	1.664
3000	1.669

6 Energy consumption per person

Our World In Data does not have data on energy usage per capita from before the year 1965, so I will be extrapolating from data on total global energy use and total population, but regardless the data is still presented on the chart below in red. All converted from kilowatt-hours per year to watts.

In green, an exponential function extrapolated from population and global energy trends:

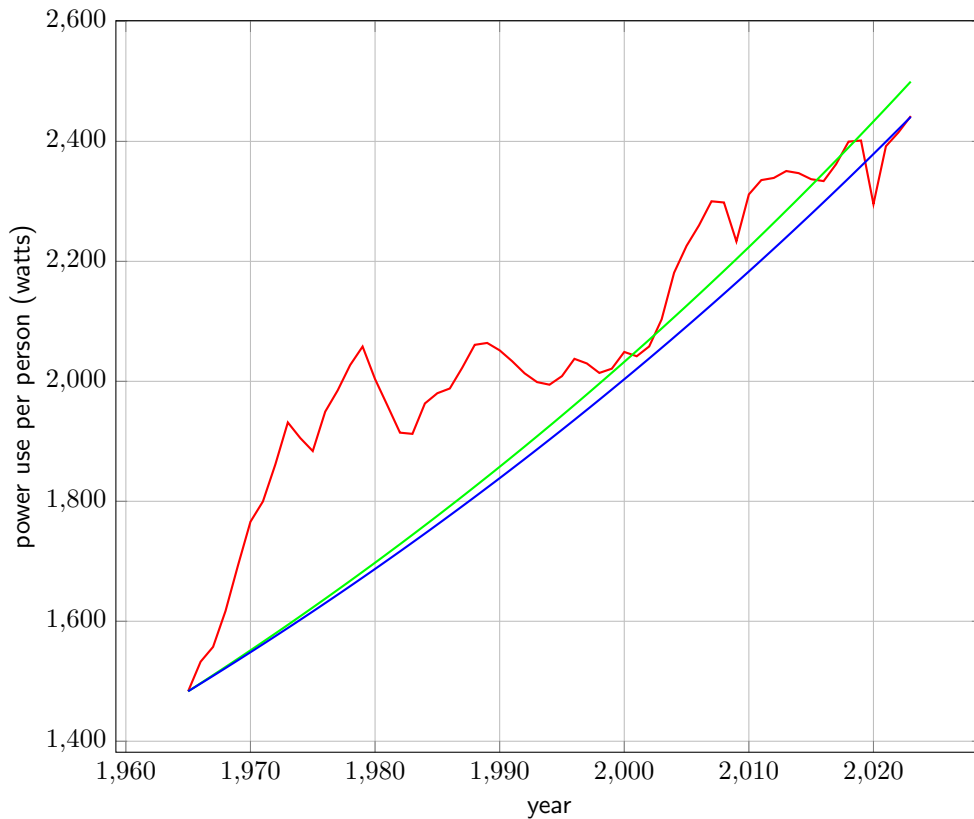
$$y = 1483.29691780822 \times 1.00904943^{x-1965}$$

Which represents a starting point at year 1965 with 1483.29691780822 watts per person, and growth of roughly 0.905% per year.

In blue shown is an exponential function calculated from data between years 1965 and 2023:

$$y = 1483.29691780822 \times 1.008634196^{x-1965}$$

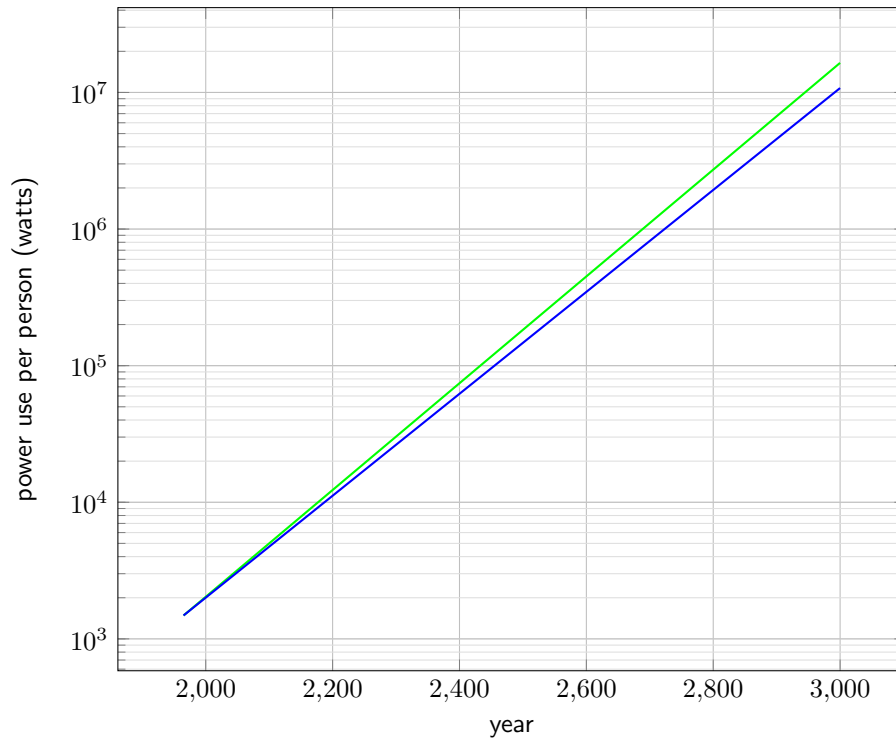
Which represents a starting point at year 1965 with 1483.29691780822 watts per person, and growth of roughly 0.863% per year.



It's noticeable that the real data shows growth, but it is not clear of what kind, personally from eyeballing it I would say maybe even logarhythmic rather than exponential, but this document focusses on exponential growth.

It is important to note though that 0.905% growth is more consistent with the growth I presented earlier for global energy use and total population size.

Extending the growth trend all the way to the year 3000,
again 0.905% in green & 0.863% in blue:



Here are some exact numbers to be extracted from this:

0.905% annual growth		
year	exact	rough
2000	2033 W	2.03×10^3 W
2050	3190 W	3.19×10^3 W
2100	5005 W	5.01×10^3 W
2150	7853 W	7.85×10^3 W
2200	12321 W	1.23×10^4 W
2250	19332 W	1.93×10^4 W
2300	30332 W	3.03×10^4 W
2350	47590 W	4.76×10^4 W
2400	74669 W	7.47×10^4 W
2450	117155 W	1.17×10^5 W
2500	183816 W	1.84×10^5 W
2600	452510 W	4.53×10^5 W
2700	1113968 W	1.11×10^6 W
2800	2742311 W	2.74×10^6 W
2900	6750887 W	6.75×10^6 W
3000	16619003 W	1.66×10^7 W

0.863% annual growth		
year	exact	rough
2000	2004 W	2.00×10^3 W
2050	3080 W	3.08×10^3 W
2100	4735 W	4.73×10^3 W
2150	7277 W	7.28×10^3 W
2200	11185 W	1.12×10^4 W
2250	17192 W	1.72×10^4 W
2300	26425 W	2.64×10^4 W
2350	40616 W	4.06×10^4 W
2400	62428 W	6.24×10^4 W
2450	95955 W	9.60×10^4 W
2500	147486 W	1.47×10^5 W
2600	348434 W	3.48×10^5 W
2700	823169 W	8.23×10^5 W
2800	1944722 W	1.94×10^6 W
2900	4594375 W	4.59×10^6 W
3000	10854136 W	1.09×10^7 W

year	W (0.905%)
2025	2.5467E+03
2030	2.6640E+03
2035	2.7868E+03
2040	2.9152E+03
2045	3.0495E+03
2050	3.1900E+03
2055	3.3369E+03
2060	3.4907E+03
2065	3.6515E+03
2070	3.8197E+03
2075	3.9957E+03
2080	4.1798E+03
2085	4.3724E+03
2090	4.5739E+03
2095	4.7846E+03
2100	5.0050E+03
2105	5.2356E+03
2110	5.4769E+03
2115	5.7292E+03
2120	5.9932E+03
2125	6.2693E+03
2130	6.5581E+03
2135	6.8603E+03
2140	7.1764E+03
2145	7.5070E+03
2150	7.8529E+03
2155	8.2147E+03
2160	8.5932E+03
2165	8.9891E+03
2170	9.4033E+03
2175	9.8365E+03
2180	1.0290E+04
2185	1.0764E+04
2190	1.1260E+04
2195	1.1778E+04
2200	1.2321E+04
2205	1.2889E+04
2210	1.3483E+04
2215	1.4104E+04
2220	1.4754E+04
2225	1.5433E+04
2230	1.6145E+04
2235	1.6888E+04
2240	1.7666E+04
2245	1.8480E+04
2250	1.9332E+04
2255	2.0223E+04
2260	2.1154E+04
2265	2.2129E+04

year	W (0.905%)
2270	2.3148E+04
2275	2.4215E+04
2280	2.5331E+04
2285	2.6498E+04
2290	2.7719E+04
2295	2.8996E+04
2300	3.0332E+04
2305	3.1729E+04
2310	3.3191E+04
2315	3.4720E+04
2320	3.6320E+04
2325	3.7993E+04
2330	3.9744E+04
2335	4.1575E+04
2340	4.3490E+04
2345	4.5494E+04
2350	4.7590E+04
2355	4.9783E+04
2360	5.2077E+04
2365	5.4476E+04
2370	5.6986E+04
2375	5.9611E+04
2380	6.2358E+04
2385	6.5231E+04
2390	6.8236E+04
2395	7.1380E+04
2400	7.4669E+04
2405	7.8109E+04
2410	8.1708E+04
2415	8.5473E+04
2420	8.9411E+04
2425	9.3530E+04
2430	9.7839E+04
2435	1.0235E+05
2440	1.0706E+05
2445	1.1200E+05
2450	1.1716E+05
2455	1.2255E+05
2460	1.2820E+05
2465	1.3411E+05
2470	1.4028E+05
2475	1.4675E+05
2480	1.5351E+05
2485	1.6058E+05
2490	1.6798E+05
2495	1.7572E+05
2500	1.8382E+05
2505	1.9229E+05
2510	2.0114E+05

year	W (0.905%)
2515	2.1041E+05
2520	2.2011E+05
2525	2.3025E+05
2530	2.4086E+05
2535	2.5195E+05
2540	2.6356E+05
2545	2.7570E+05
2550	2.8841E+05
2555	3.0170E+05
2560	3.1560E+05
2565	3.3014E+05
2570	3.4535E+05
2575	3.6126E+05
2580	3.7790E+05
2585	3.9531E+05
2590	4.1353E+05
2595	4.3258E+05
2600	4.5251E+05
2605	4.7336E+05
2610	4.9517E+05
2615	5.1798E+05
2620	5.4185E+05
2625	5.6681E+05
2630	5.9293E+05
2635	6.2025E+05
2640	6.4882E+05
2645	6.7872E+05
2650	7.0999E+05
2655	7.4270E+05
2660	7.7692E+05
2665	8.1271E+05
2670	8.5016E+05
2675	8.8933E+05
2680	9.3030E+05
2685	9.7316E+05
2690	1.0180E+06
2695	1.0649E+06
2700	1.1140E+06
2705	1.1653E+06
2710	1.2190E+06
2715	1.2751E+06
2720	1.3339E+06
2725	1.3954E+06
2730	1.4596E+06
2735	1.5269E+06
2740	1.5972E+06
2745	1.6708E+06
2750	1.7478E+06
2755	1.8283E+06

year	W (0.905%)
2760	1.9126E+06
2765	2.0007E+06
2770	2.0929E+06
2775	2.1893E+06
2780	2.2902E+06
2785	2.3957E+06
2790	2.5061E+06
2795	2.6215E+06
2800	2.7423E+06
2805	2.8687E+06
2810	3.0008E+06
2815	3.1391E+06
2820	3.2837E+06
2825	3.4350E+06
2830	3.5933E+06
2835	3.7588E+06
2840	3.9320E+06
2845	4.1132E+06
2850	4.3027E+06
2855	4.5009E+06
2860	4.7083E+06
2865	4.9252E+06
2870	5.1521E+06
2875	5.3895E+06
2880	5.6378E+06
2885	5.8976E+06
2890	6.1693E+06
2895	6.4535E+06
2900	6.7509E+06
2905	7.0619E+06
2910	7.3873E+06
2915	7.7276E+06
2920	8.0837E+06
2925	8.4561E+06
2930	8.8457E+06
2935	9.2533E+06
2940	9.6796E+06
2945	1.0126E+07
2950	1.0592E+07
2955	1.1080E+07
2960	1.1591E+07
2965	1.2125E+07
2970	1.2683E+07
2975	1.3268E+07
2980	1.3879E+07
2985	1.4518E+07
2990	1.5187E+07
2995	1.5887E+07
3000	1.6619E+07

year	W (0.863%)
2025	2.4846E+03
2030	2.5937E+03
2035	2.7076E+03
2040	2.8265E+03
2045	2.9507E+03
2050	3.0803E+03
2055	3.2156E+03
2060	3.3568E+03
2065	3.5043E+03
2070	3.6582E+03
2075	3.8189E+03
2080	3.9866E+03
2085	4.1617E+03
2090	4.3445E+03
2095	4.5353E+03
2100	4.7345E+03
2105	4.9425E+03
2110	5.1596E+03
2115	5.3862E+03
2120	5.6228E+03
2125	5.8697E+03
2130	6.1275E+03
2135	6.3967E+03
2140	6.6776E+03
2145	6.9709E+03
2150	7.2771E+03
2155	7.5968E+03
2160	7.9304E+03
2165	8.2788E+03
2170	8.6424E+03
2175	9.0220E+03
2180	9.4183E+03
2185	9.8319E+03
2190	1.0264E+04
2195	1.0715E+04
2200	1.1185E+04
2205	1.1677E+04
2210	1.2189E+04
2215	1.2725E+04
2220	1.3284E+04
2225	1.3867E+04
2230	1.4476E+04
2235	1.5112E+04
2240	1.5776E+04
2245	1.6469E+04
2250	1.7192E+04
2255	1.7947E+04
2260	1.8736E+04
2265	1.9558E+04

year	W (0.863%)
2270	2.0418E+04
2275	2.1314E+04
2280	2.2251E+04
2285	2.3228E+04
2290	2.4248E+04
2295	2.5313E+04
2300	2.6425E+04
2305	2.7586E+04
2310	2.8797E+04
2315	3.0062E+04
2320	3.1382E+04
2325	3.2761E+04
2330	3.4200E+04
2335	3.5702E+04
2340	3.7270E+04
2345	3.8907E+04
2350	4.0616E+04
2355	4.2400E+04
2360	4.4262E+04
2365	4.6207E+04
2370	4.8236E+04
2375	5.0355E+04
2380	5.2566E+04
2385	5.4875E+04
2390	5.7286E+04
2395	5.9802E+04
2400	6.2428E+04
2405	6.5170E+04
2410	6.8033E+04
2415	7.1021E+04
2420	7.4141E+04
2425	7.7397E+04
2430	8.0797E+04
2435	8.4345E+04
2440	8.8050E+04
2445	9.1918E+04
2450	9.5955E+04
2455	1.0017E+05
2460	1.0457E+05
2465	1.0916E+05
2470	1.1396E+05
2475	1.1896E+05
2480	1.2419E+05
2485	1.2964E+05
2490	1.3534E+05
2495	1.4128E+05
2500	1.4749E+05
2505	1.5396E+05
2510	1.6073E+05

year	W (0.863%)
2515	1.6779E+05
2520	1.7516E+05
2525	1.8285E+05
2530	1.9088E+05
2535	1.9926E+05
2540	2.0802E+05
2545	2.1715E+05
2550	2.2669E+05
2555	2.3665E+05
2560	2.4704E+05
2565	2.5789E+05
2570	2.6922E+05
2575	2.8105E+05
2580	2.9339E+05
2585	3.0628E+05
2590	3.1973E+05
2595	3.3377E+05
2600	3.4843E+05
2605	3.6374E+05
2610	3.7971E+05
2615	3.9639E+05
2620	4.1380E+05
2625	4.3198E+05
2630	4.5095E+05
2635	4.7076E+05
2640	4.9144E+05
2645	5.1302E+05
2650	5.3556E+05
2655	5.5908E+05
2660	5.8363E+05
2665	6.0927E+05
2670	6.3603E+05
2675	6.6397E+05
2680	6.9313E+05
2685	7.2357E+05
2690	7.5536E+05
2695	7.8853E+05
2700	8.2317E+05
2705	8.5932E+05
2710	8.9707E+05
2715	9.3647E+05
2720	9.7760E+05
2725	1.0205E+06
2730	1.0654E+06
2735	1.1122E+06
2740	1.1610E+06
2745	1.2120E+06
2750	1.2652E+06
2755	1.3208E+06

year	W (0.863%)
2760	1.3788E+06
2765	1.4394E+06
2770	1.5026E+06
2775	1.5686E+06
2780	1.6375E+06
2785	1.7094E+06
2790	1.7845E+06
2795	1.8629E+06
2800	1.9447E+06
2805	2.0301E+06
2810	2.1193E+06
2815	2.2124E+06
2820	2.3096E+06
2825	2.4110E+06
2830	2.5169E+06
2835	2.6275E+06
2840	2.7429E+06
2845	2.8633E+06
2850	2.9891E+06
2855	3.1204E+06
2860	3.2575E+06
2865	3.4005E+06
2870	3.5499E+06
2875	3.7058E+06
2880	3.8686E+06
2885	4.0385E+06
2890	4.2159E+06
2895	4.4011E+06
2900	4.5944E+06
2905	4.7962E+06
2910	5.0068E+06
2915	5.2268E+06
2920	5.4563E+06
2925	5.6960E+06
2930	5.9462E+06
2935	6.2073E+06
2940	6.4800E+06
2945	6.7646E+06
2950	7.0617E+06
2955	7.3719E+06
2960	7.6957E+06
2965	8.0337E+06
2970	8.3866E+06
2975	8.7549E+06
2980	9.1395E+06
2985	9.5409E+06
2990	9.9600E+06
2995	1.0397E+07
3000	1.0854E+07